# Database Design Process

Lecture Notes · Game DB Junior

**In this lesson**

- The **3 questions** we ask to design a database
- What a **table**, a **column** and a **row** are
- How to plan a **cities** table from a Wikipedia page

## 1. What we focus on

There are four different challenges when we work with databases. In this part we focus on the first two:

- How to write **efficient queries**.
- How to design the **schema** (the structure) of our database.

Our first small project with Postgres: store a **list of large cities** in a database.

## 2. The data source

The data comes from the Wikipedia page *List of largest cities*. The page has a table with many columns, but we only care about **four** of them:

| Column on Wikipedia | What it means |
|---|---|
| City | The name of the city |
| Country | The country of the city |
| Population (urban area) | How many people live in the urban area around the city |
| Area (urban area) | The size of the urban area around the city |

## 3. The design process: 3 questions

We will use this small design process many times in the course. We ask ourselves three questions, and we use the answers to set up our database.

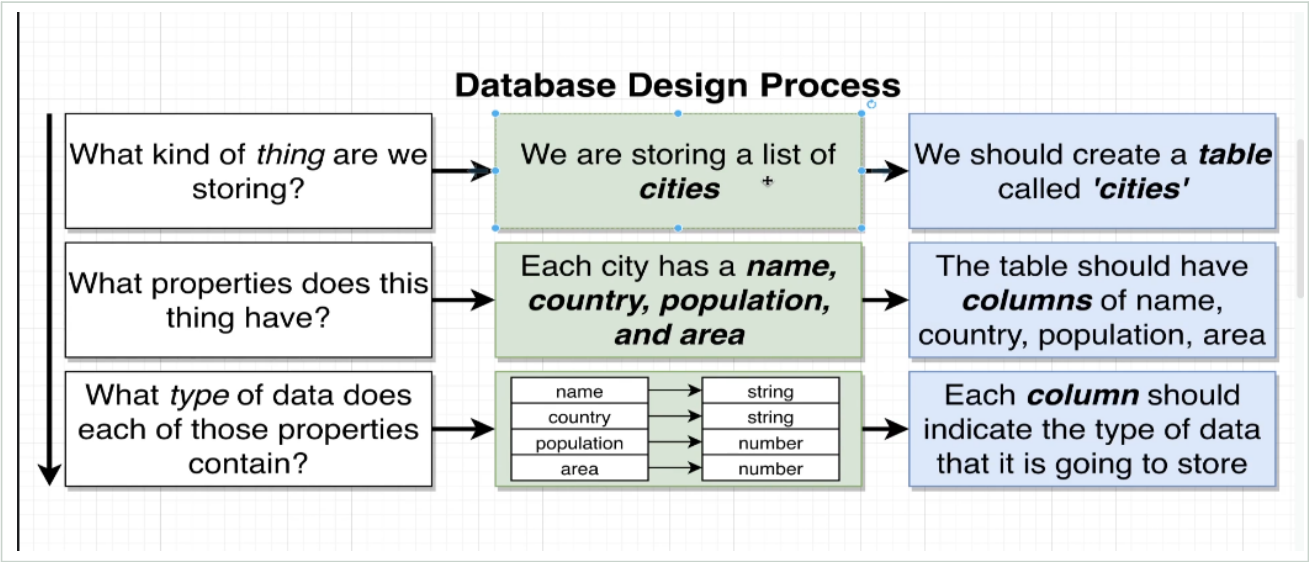



*Figure 1 — Database Design Process*

| Question | Answer for cities | What we do in the database |
|---|---|---|
| 1. What kind of *thing* are we storing? | A list of cities | Create a **table** called **cities** |
| 2. What properties does this thing have? | name, country, population, area | Create **columns**: name, country, population, area |
| 3. What type of data does each property contain? | name, country → string<br>population, area → number | Each column says what **type of data** it will store |

## 4. Table

The answer to question 1 is "a list of cities". So we create a **table** in our database and we name it **cities**.

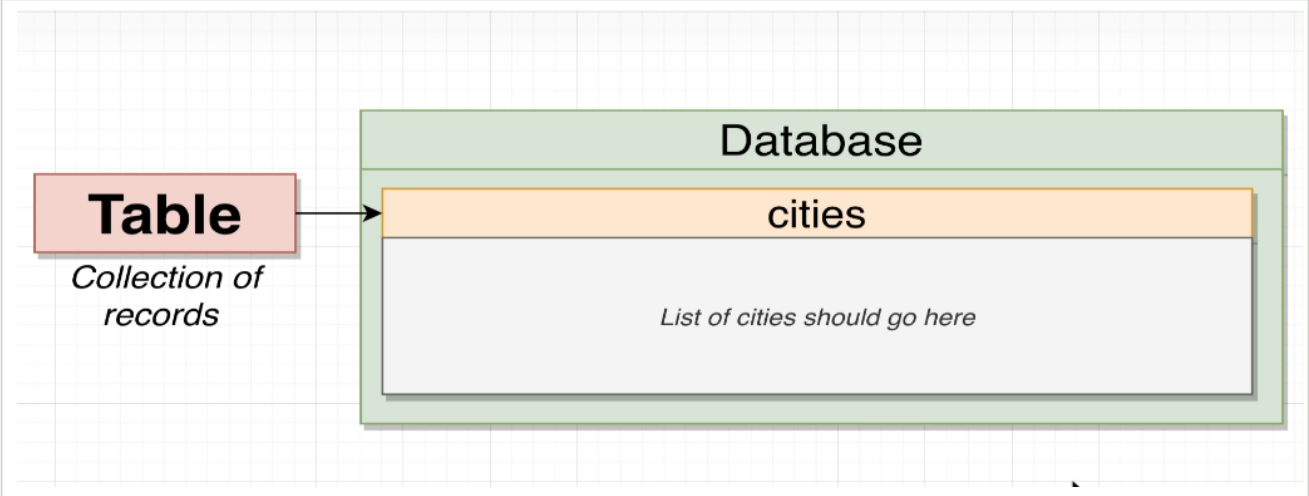



*Figure 2 — A table is a collection of records*

- A table sits inside the database.
- A table stores a **collection of records** that are related to each other.
- Later, we will put all the cities from the Wikipedia page into the cities table.

- We **only** store data inside tables. So it is very important to think about how to name and design our tables.

## 5. Columns

The answer to question 2 gives us the properties of a city. We use each property to make a **column** in the table. We want to store the name, country, population and area of each city, so we make **four columns**, one for each property.

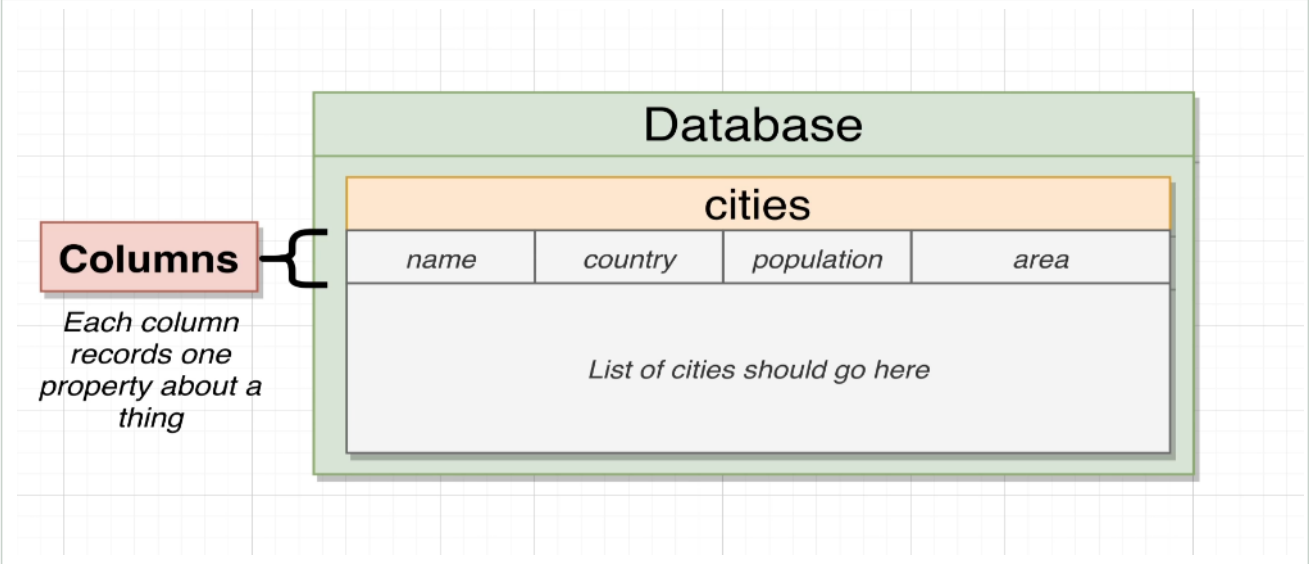



*Figure 3 — Each column records one property about a thing*

## 6. Type of data

The answer to question 3 tells us what **type of data** each column will store:

| Column | Type of data |
|---|---|
| name | string |
| country | string |
| population | number |
| area | number |

We will see where we use this information in the next lesson.

## 7. Rows

After we set up the table and its four columns, we can add data into it. For example, we add Tokyo, Delhi and Shanghai. Each city has a name, a country, a population and an area.

Each city that we add into the table is called a **row**. Right now the cities table has **three rows**: one row is Tokyo, one is Delhi, one is Shanghai.

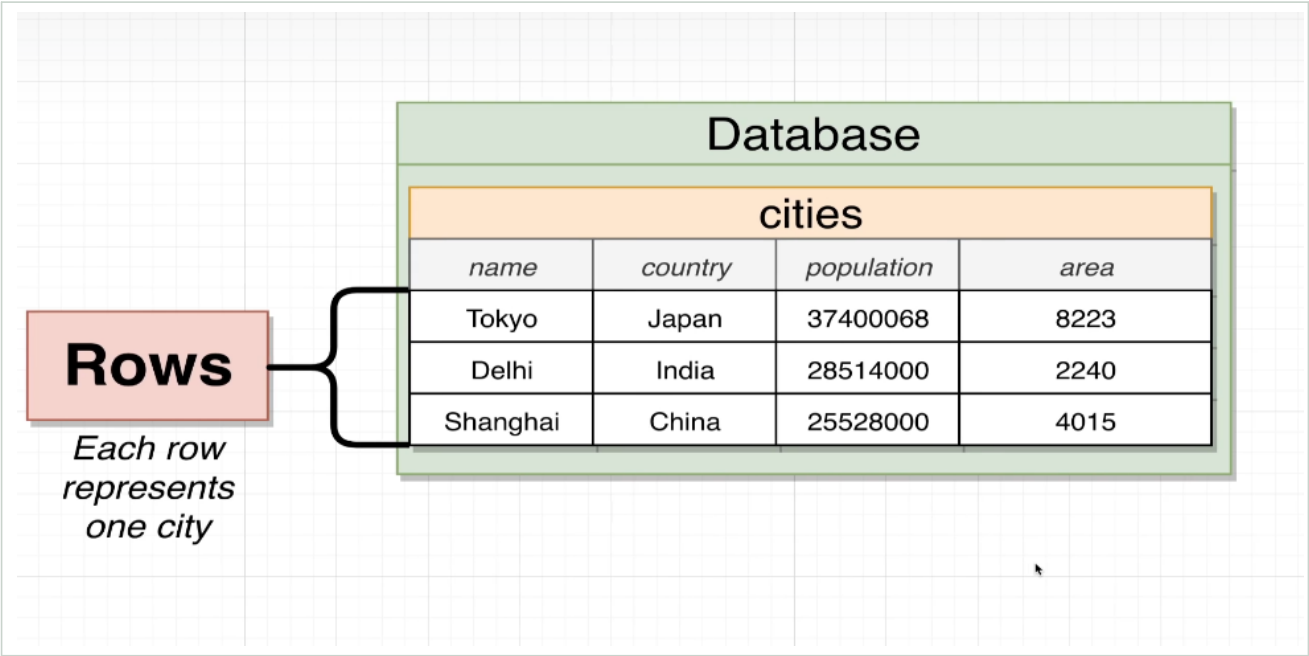



*Figure 4 — Each row represents one city*

### Numbers on Wikipedia today

The Wikipedia page has changed since the lesson was recorded. Here are the lesson numbers and the numbers on the page today (urban area):

| City | Country | Population (lesson) | Area (lesson) | Population (Wikipedia today) | Area (Wikipedia today) |
|---|---|---|---|---|---|
| Tokyo | Japan | 37400068 | 8223 | 37785000 | 8231 |
| Delhi | India | 28514000 | 2240 | 32226000 | 2344 |
| Shanghai | China | 25528000 | 4015 | 24073000 | 4333 |

Source: Wikipedia, "List of largest cities", urban area columns. Checked on 27 October 2026.

## 8. Summary

| Word | Meaning | Example |
|---|---|---|
| Table | A collection of records that are related | cities |
| Column | Stores one property about a thing | name, country, population, area |
| Row | One record inside a table | Tokyo \| Japan \| 37400068 \| 8223 |
| Type of data | What kind of data a column stores | string, number |

**Remember:** Tables have many columns. Each column stores one property. Each record inside a table is a row.

Next lesson: we start to build the cities table.

**Sources**

Lecture transcript and lecture images (Figures 1–4)
Wikipedia — List of largest cities:

https://en.wikipedia.org/wiki/List_of_largest_cities